Physical Chemistry Cume

1. The vapor pressure of white phosphorus at 69° C temperature (342.2 K) is 100 Pa (≈0.75 mm Hg), and the vapor density (at 69° C) is 4.355 mg/L.
Based on this, what is the molecular formula of phosphorus in the gas phase at 69°C? (E.g., P, P_2, P_3, P_4, P_5, P_6, P_8 … ?) Assume that the phosphorus vapor is an ideal gas. R=8.3145 J K^{-1} mol^{-1}. The atomic mass of phosphorus is 30.974 g/mol.
(Help: 1 J = 1 Pa*1 m^3 = 1000 Pa*1 L.) (50 p)

2. How high must a waterfall be for the tailwater to be 0.1 K warmer than the headwater?
The gravitational energy of the water at the top of the waterfall is first converted to kinetic energy and finally to heat; essentially all the energy of the fall goes into warming the water. The specific (per kg) heat capacity of water is C_p = 4.184 kJ/(K kg), the gravitational acceleration is g = 9.81 m s^{-2}. [Help: 1 J = 1 kg $m^{2\,s}$-2. Be sure to notice that C_P is given in kJ, not J, units.] (50 p)